

Chapter 14: Azure DevOps

Learning Objectives

By the end of this chapter, you will:

- Understand the core capabilities of Azure DevOps and its role in enterprise AI delivery.
- Learn how AI TPMs use Azure DevOps for planning, tracking, CI/CD, and reporting.
- Develop an understanding of modern DevOps practices for AI applications and services.

Beginner Explanation

Azure DevOps is Microsoft's platform for planning, building, testing, and deploying software.

Think of Azure DevOps as the "Operating System" for a Technical Program Manager.

It provides tools for:

- Planning work
- Managing backlogs
- Tracking sprints
- Building pipelines
- Deploying applications
- Monitoring releases

Azure DevOps Components

Service	Purpose
Boards	Manage work items
Repos	Store source code
Pipelines	Automate builds and deployments
Test Plans	Manage testing activities
Artifacts	Store packages and dependencies

Example

Suppose your team is building an AI Assistant.

Azure DevOps helps answer:

- What work remains?
- Who owns each task?
- Which sprint are we in?
- Has the application been deployed?

- Are there any blockers?

For AI TPMs, Azure DevOps becomes the single source of truth for delivery.

Intermediate Explanation

Enterprise AI programs typically involve multiple teams:

- Product
- Engineering
- Data Science
- Security
- Infrastructure
- QA
- Operations

Azure DevOps helps coordinate work across all these teams.

AI Delivery Lifecycle

Phase	Azure DevOps Capability
Planning	Boards
Development	Repos
Build	Pipelines
Testing	Test Plans
Deployment	Release Pipelines
Monitoring	Dashboards

Typical Work Item Hierarchy

``text id="5e9wxn" Epic ↓ Feature ↓ User Story ↓ Task ↓ Bug

Example

```
| Level | Example |
|-----|-----|
| Epic | Enterprise AI Assistant |
| Feature | HR Policy Search |
| User Story | Search PTO Policies |
| Task | Build Search Index |
| Bug | Incorrect Search Results |
```

CI/CD for AI Applications

Modern AI applications require:

- Infrastructure deployment
- Prompt deployment
- Model configuration
- Testing
- Monitoring

AI TPMs should understand how CI/CD pipelines support these activities.

Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Provides visibility into program status and delivery progress.
Risk	Helps identify blockers, delays, and dependencies.
Stakeholder Management	Enables transparent reporting through dashboards.
Budget	Supports resource planning and utilization tracking.

Enterprise Example

Business Problem

A global insurance company managed more than 20 AI initiatives across multiple regions.

Challenges included:

- Lack of visibility
- Manual status reporting
- Inconsistent deployments
- Missed deadlines

Solution

Standardize delivery using Azure DevOps.

Architecture

- Azure Boards
- Azure Repos
- Azure Pipelines
- Azure Test Plans
- Azure Dashboards
- Azure OpenAI

Outcome

Metric	Result
Delivery Predictability	Increased by 32%
Deployment Time	Reduced by 60%
Release Failures	Reduced by 45%
Stakeholder Satisfaction	94%

Lessons Learned

- Dashboards significantly improve visibility.
- Standardized pipelines reduce operational issues.
- Sprint metrics help identify bottlenecks.
- Automation accelerates delivery.

Azure Implementation

Services Used

Service	Purpose
Azure DevOps	Delivery management
Azure OpenAI	AI application layer
Azure Functions	Workflow orchestration
Azure AI Search	Enterprise retrieval

Implementation Workflow

Step 1: Configure Azure Boards

Create:

- Epics
- Features
- User Stories
- Tasks
- Bugs

Step 2: Set Up Repositories

Store:

- Application code
- Infrastructure as Code
- Prompt templates

- Documentation

Step 3: Configure Pipelines

Typical pipeline stages:

1. Build
2. Test
3. Security Scan
4. Deploy
5. Validate

Step 4: Enable Dashboards

Track:

- Sprint velocity
- Open defects
- Deployment frequency
- Lead time
- Risks

Step 5: Conduct Reviews

Review:

- Sprint metrics
- Delivery KPIs
- Costs
- Risks
- Team performance

Mermaid Diagram

```
```mermaid id="vhmq2"  
graph TD
```

```
A[Azure Boards]
```

```
A --> B[Azure Repos]
```

```
B --> C[Azure Pipelines]
```

```
C --> D[Azure Test Plans]
D --> E[Production Deployment]
E --> F[Dashboards]
F --> G[Stakeholders]
```

---

## AI CI/CD Pipeline

```
```mermaid id="qajk6n" graph LR
```

```
A[Code Commit]
```

```
A --> B[Build]
```

```
B --> C[Test]
```

```
C --> D[Security Scan]
```

```
D --> E[Deploy]
```

```
E --> F[Azure OpenAI]
```

```
F --> G[Production] ```
```

TPM Checklist

- ☐ Define work item hierarchy.
 - ☐ Configure sprint boards.
 - ☐ Establish CI/CD pipelines.
 - ☐ Define KPIs.
 - ☐ Create executive dashboards.
 - ☐ Conduct sprint reviews.
 - ☐ Review deployment metrics.
 - ☐ Track risks and blockers.
 - ☐ Monitor delivery trends.
 - ☐ Obtain stakeholder feedback.
-

Templates

Sprint Dashboard

Metric	Target
Velocity	Increasing
Sprint Completion	>90%
Defect Leakage	<2%
Deployment Success	>95%
Team Satisfaction	>90%

RAID Log

Category	Description
Risks	
Assumptions	
Issues	
Dependencies	

Executive Dashboard

KPI	Status
Delivery	Green
Costs	Green
Risks	Yellow
Quality	Green
Adoption	Green

Common Mistakes

1. Treating Azure DevOps as only a ticketing tool.
2. Maintaining inconsistent work item structures.
3. Ignoring sprint metrics.
4. Skipping automation opportunities.
5. Failing to maintain dashboards.
6. Neglecting dependency tracking.

7. Not standardizing CI/CD pipelines.

Key Takeaways

- Azure DevOps is a critical platform for managing enterprise AI delivery.
 - AI TPMs should understand Boards, Repos, Pipelines, and Dashboards.
 - CI/CD pipelines improve reliability and reduce deployment risks.
 - Dashboards provide transparency for stakeholders and executives.
 - Standardized delivery processes improve predictability and team performance.
 - Azure DevOps enables organizations to scale AI initiatives efficiently.
-

Footer

charan.ai | Page 67